The Water Cycle

Why water keeps moving

Every drop of water on Earth has a long history. The water you drink today may once have been part of a cloud, a river, a glacier, or even the sweat of a dinosaur. That is possible because Earth does not make new water or lose much of its old water. Instead, the same water travels again and again between the oceans, the air, the land, and living things. This endless journey is called the water cycle.

The water cycle is powered by two forces. The first is the Sun, which supplies heat. The second is gravity, which pulls water downward. Heat lifts water into the air, and gravity brings it back to the ground. Together they keep water moving all the time, day and night, in every part of the world.

Scientists often describe the water cycle in four main stages: evaporation, condensation, precipitation, and collection. A fifth process, transpiration, happens in plants and adds a lot of water vapor to the air.

Evaporation

Evaporation is the change of liquid water into water vapor, which is a gas. When the Sun warms the surface of an ocean, a lake, or a puddle, some of the water molecules move faster and faster. Eventually, the fastest molecules escape from the surface and rise into the air as an invisible gas.

Condensation

As water vapor rises, it cools. Cool air cannot hold as much vapor as warm air, so the vapor turns back into tiny droplets of liquid water around specks of dust. Billions of these droplets together form a cloud.

Precipitation and collection

When the droplets in a cloud join and grow too heavy to float, gravity pulls them down as rain, snow, sleet or hail. The water collects in oceans, lakes, rivers and underground, and the cycle begins again.